

Demographic Narcolepsy Data Study

February 7, 2023

```
[1]: import numpy as np
import pandas as pd
import scipy as sp
import matplotlib as plt
import seaborn as sns
from scipy.stats import chi2_contingency
import matplotlib.pyplot as plt

TypeData = pd.read_csv('Type Sheet.csv')
```

Main Question: What phenotypic characteristics distinguish Type 1 and Type 2 narcolepsy within the dataset?

Supplementary Questions: 1) Are there any noticeable patterns between sex and symptoms for one narcolepsy type? 2) Are there any noticeable patterns between younger and older age group and symptoms? 3) Are there any significant relationships between narcolepsy type and one symptom? 4) Are there any significant correlations between two symptoms in Group 1 that could be attributed to sex rather than type?

Note: This file only shows the significant findings from my data manipulations due to the large amount of code. I will include the rest of my findings in the zip file.

Are there any noticeable patterns between sex and symptoms for each narcolepsy type?

The following outputs in this section are: 1) bar graph showing average occurrence of categorical narcolepsy symptoms for each sex and narcolepsy type 2) grid histograms of SOREM periods for each sex and narcolepsy type 3) grid histograms of sleep latency for each sex and narcolepsy type

Results: In the categorical data (bar graph), there appears to be major differences between sex regarding cataplexy. In Type 1 participants, there is a greater average of cataplexy in males than females. In Type 2 participants, there is a greater average of cataplexy in females than males. Also, Type 2 females experience cataplexy much more than Type 1 females, and Type 1 males experience cataplexy more than Type 2 males.

Females display a possible difference in REM behaviour disorders (RBDs) depending on narcolepsy type. The average for Type 1 females experiencing RBDs is less than Type 2 females experiencing RBDs, which is not shown in males. Additionally, Type 1 males experience more lucid dreaming on average than Type 2 males by a large amount, which is not seen in females. Hallucinations and sleep paralysis do not appear to “favor” one sex as there is not a major difference in averages.

Regarding SOREM periods, there does appear to be a distribution difference between Type 1 males and females. Type 1 males have a surprisingly normal distribution across SOREM periods whereas Type 1 females appear to be less normally distributed with the greatest number experiencing 3 SOREM periods. Both Type 1 male and Type 1 female distribution are skewed to the left, which makes sense considering how narcolepsy diagnosis works. Patients with sleep disorders indicating narcolepsy are required to take an MSLT. The MSLT involves taking 5 twenty-minute naps during the day while brain sleep waves are recorded. If REM waves are found during at least 2 naps (SOREM periods), then the patient passes the major criterion of the narcolepsy diagnosis. Most confirmed narcolepsy patients experience over 3 SOREM periods during the MSLT. In Type 2 males and females, there is no normal distribution, and both experience 5 SOREM periods.

The distributions of sleep latency between sex and type do not appear significantly different.

Interpretation: Type 2 narcolepsy is typically categorized as narcolepsy without cataplexy. So, it is surprising how many Type 2 females have cataplexy compared to Type 1 females. It is also intriguing that the Type 1 males experience cataplexy more than Type 1 females. Considering possible error in data collection for Type 2 females experiencing cataplexy, there is most certainly a sex difference in Type 1 patients regarding cataplexy.

The greater average difference in females with RBDs could be attributed to the population from which the data was collected, especially since there is little difference between females in the other related REM conditions (hallucinations and sleep paralysis). However, the major difference in average males with lucid dreaming between types is intriguing because it is not fully understood by the scientific community.

For SOREM periods, it's interesting that Type 1 males have a normal distribution with a median of 4 SOREM periods and that both Type 1 sexes are more normally distributed than Type 2. It is thought that Type 1 narcolepsy is heritable while Type 2 is caused by environmental factors. This idea has not been proven yet. The greater incidence of permanent SOREM sleep (5 SOREMS) in Type 2 patients could support the idea that a sudden environmental change “flipped a neural switch” while the even distribution of Type 1 SOREM periods could support the manifestation of multiple genetic factors. However, this is just a guess and would need to be supported by much more data.

```
[2]: #to easily examine the categorical variables (Cataplexy, hallucinations, sleep_
    ↪paralysis, RBD, lucid dreaming)
    #a bar graph can be made of the mean occurrence for each gender and type

    #isolate the narcolepsy group and save in namespace
    group1type = TypeData[(TypeData['Group'] == 1)]

    #isolate the Type 1 group using the groupby function, aggregate using the mean_
    ↪function, save in namespace
    type1means = group1type[(group1type['Narcolepsy Type'] == '1')].groupby('Sex').
    ↪mean()
    #reset the index for later concatenation and save in namespace
    type1_means = type1means.reset_index()
```

```

#isolate the Type 2 group using the groupby function, aggregate using the mean
↳function, save in namespace
type2means = group1type[(group1type['Narcolepsy Type'] == '2')].groupby('Sex').
↳mean()
#reset the index for later concatenation and save in namespace
type2_means = type2means.reset_index()

#for both type groups, select the similar columns and save in namespace
means1 = type1_means[['Sex', 'Cataplexy', 'Hypnagogic hallucinations', 'Sleep
↳paralysis', 'RBD (REM behaviour disorder)', 'Lucid dreaming']]
means2 = type2_means[['Sex', 'Cataplexy', 'Hypnagogic hallucinations', 'Sleep
↳paralysis', 'RBD (REM behaviour disorder)', 'Lucid dreaming']]

#merge the above group mean tables using the concat function, rename the rows
↳using the set axis function, and reset the index
newsymconcat = pd.concat((means1, means2), axis=0).set_axis(['Type 1 F', 'Type
↳1 M', 'Type 2 F', 'Type 2 M'], axis='rows', inplace=False).reset_index()

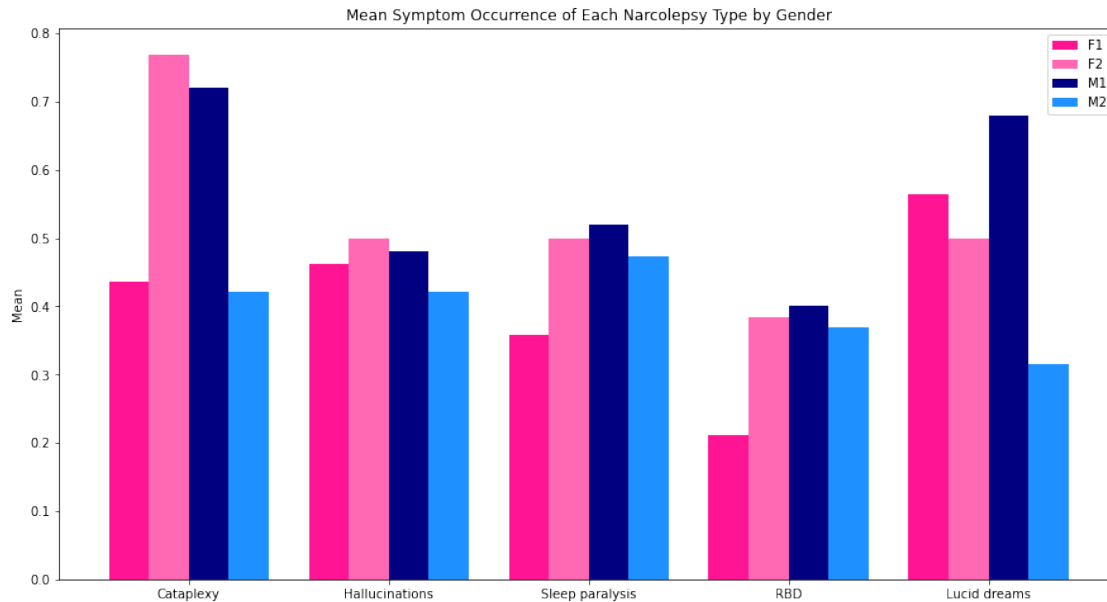
#for the new table, select the desired columns for the bar graph and save in
↳namespace
new_sym = newsymconcat[['index', 'Cataplexy', 'Hypnagogic hallucinations',
↳'Sleep paralysis', 'RBD (REM behaviour disorder)', 'Lucid dreaming']]
#rename the columns for cleaner x-axis labels
new_sym.set_axis(['Sex', 'Cataplexy', 'Hallucinations', 'Sleep paralysis',
↳'RBD', 'Lucid dreams'], axis='columns', inplace=True)

#select the desired columns, switch the columns for the rows using the
↳transpose function, save in namespace
newsym = new_sym[['Cataplexy', 'Hallucinations', 'Sleep paralysis', 'RBD',
↳'Lucid dreams']].T
#rename the columns for simpler bar graph legend
newsym.set_axis(['F1', 'M1', 'F2', 'M2'], axis='columns', inplace=True)
#set the new row values as the index using the reindex function and save the
↳final table in namespace
newsymtype = newsym.reindex(sorted(newsym.columns), axis=1)

#plot the bar graph with appropriate size, axis labelling, and colors
newsymtype.plot.bar(figsize=(15,8), rot=0, color=['deeppink', 'hotpink',
↳'navy', 'dodgerblue'], ylabel='Mean', title='Mean Symptom Occurrence of Each
↳Narcolepsy Type by Gender', width=0.8)

```

[2]: <AxesSubplot:title={'center': 'Mean Symptom Occurrence of Each Narcolepsy Type by Gender'}, ylabel='Mean'>



```
[3]: #for the numerical symptoms (SOREM periods and sleep latency), make grid
      ↪ histograms to examine multiple variables

      #in original dataset, select for desired columns and save in namespace
      sexgroupSorem = TypeData[['Sex', 'Narcolepsy Type', 'Sleep onset in REM
      ↪ periods']]

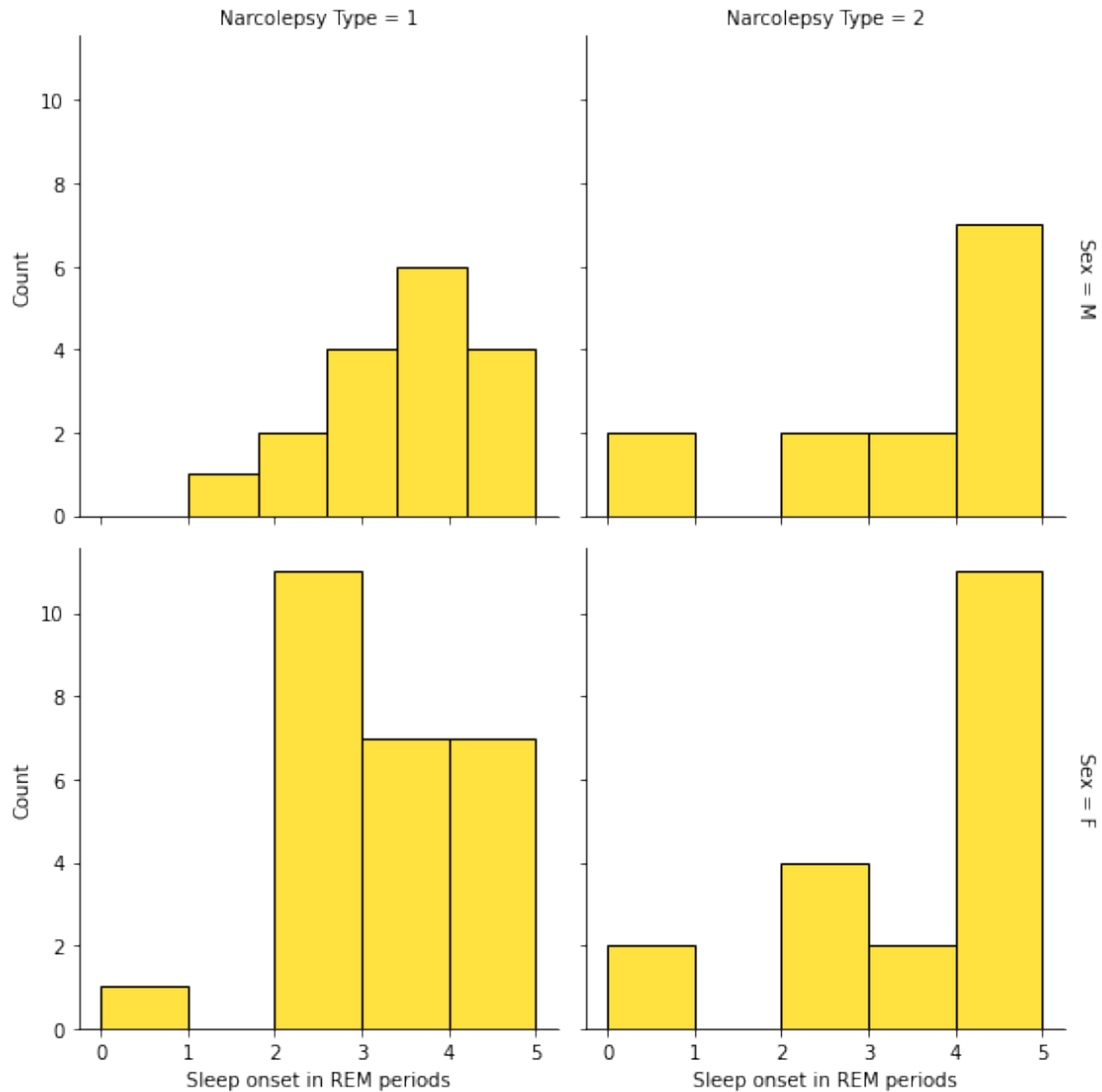
      #remove narcolepsy types that are not equal to '1' or '2'
      sgS = sexgroupSorem[(sexgroupSorem['Narcolepsy Type'] != 0) &
      ↪ (sexgroupSorem['Narcolepsy Type'] != '0') & (sexgroupSorem['Narcolepsy
      ↪ Type'] != '-')]

      #reorder to values for the narcolepsy type column so that the types appear in
      ↪ numerical order in the grid
      sgs = sgS.sort_values(['Narcolepsy Type'])

      #use seaborn to set up a graphing grid for each narcolepsy type and sex
      grid = sns.FacetGrid(data=sgs, row = 'Sex', col='Narcolepsy Type',
      ↪ margin_titles=True, height=4, aspect=1.0)

      #use seaborn to make a grid of histograms with SOREM on the x-axis
      grid.map_dataframe(sns.histplot, 'Sleep onset in REM periods', bins=5,
      ↪ color='gold')
```

```
[3]: <seaborn.axisgrid.FacetGrid at 0x7fb220855070>
```

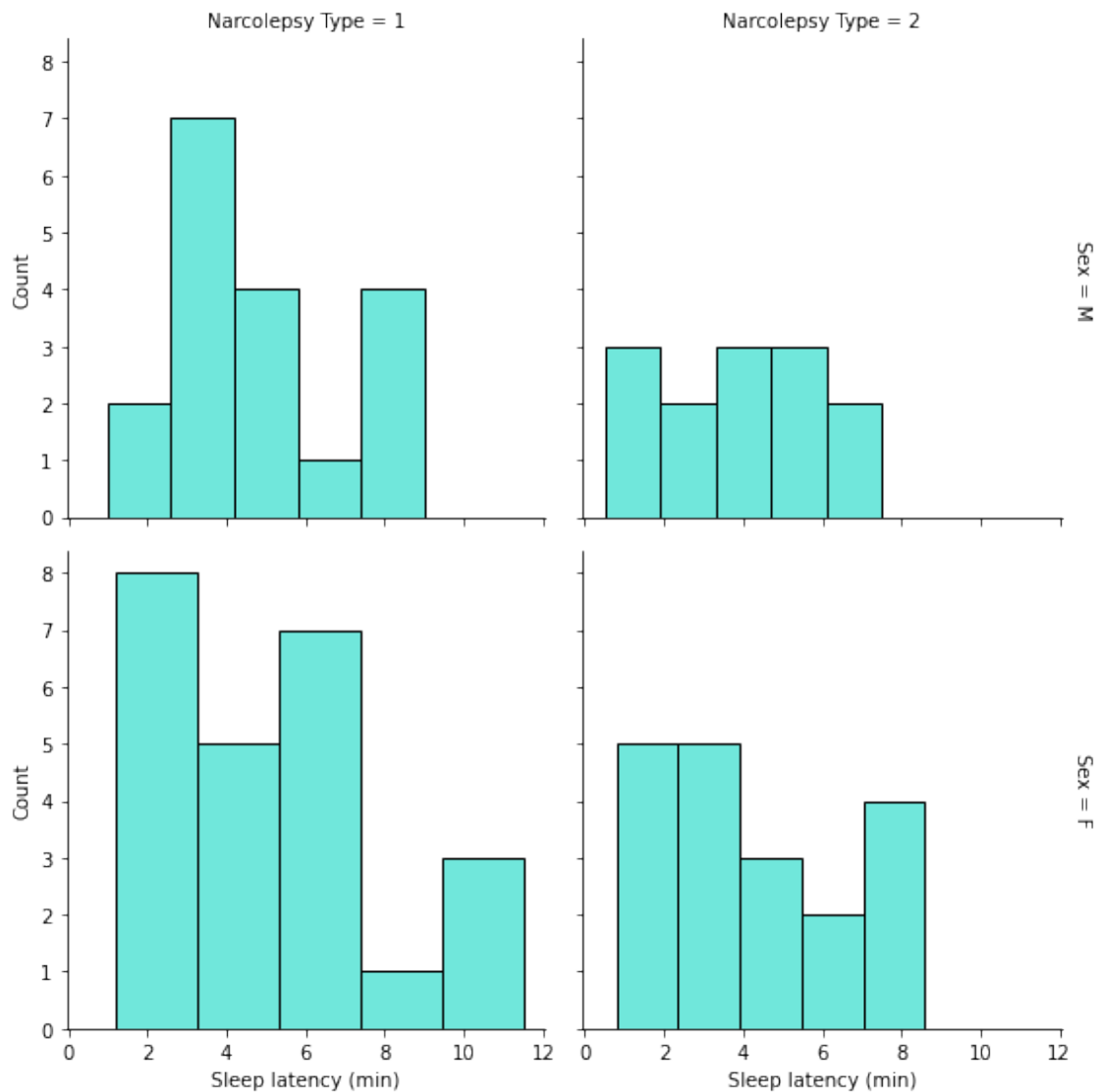


```
[4]: #in original dataset, select for desired columns and save in namespace
sextypelatency = TypeData[['Sex', 'Group', 'Narcolepsy Type', 'Sleep latency_
    ↳(min)']]
#remove narcolepsy type values that are not '1' or '2' and save in namespace
sextypelatency = sextypelatency[(sextypelatency['Narcolepsy Type'] != 0) &
    ↳(sextypelatency['Narcolepsy Type'] != '0') & (sextypelatency['Narcolepsy_
    ↳Type'] != '-')]
#reorder to values for the narcolepsy type column so that the types appear in_
    ↳numerical order in the grid
stL = sextypelatency.sort_values(['Narcolepsy Type'])

#use seaborn to set up a graphing grid for each narcolepsy type and sex
```

```
grid = sns.FacetGrid(data=stL, row='Sex', col='Narcolepsy Type',
    ↪margin_titles=True, height=4, aspect=1.0)
#use seaborn to make a grid of histograms with sleep latency on the x-axis
grid.map_dataframe(sns.histplot, 'Sleep latency (min)', bins=5,
    ↪color='turquoise')
```

[4]: <seaborn.axisgrid.FacetGrid at 0x7fb220d4a070>



Question 2 - Are there any noticeable patterns between younger and older age groups and symptoms?

The following outputs in this section are: 1) bar graph showing average occurrence of categorical narcolepsy symptoms for each age group and narcolepsy type 2) grid histograms of SOREM periods for each age group and narcolepsy type 3) scatterplots of sleep latency for each age group and

narcolepsy type

Results: There do not appear to be any major mean differences for cataplexy, hallucinations, and lucid dreaming between age group for each type. However, differences do appear for sleep paralysis, RBD, and lucid dreaming. Sleep paralysis appears on average to occur more frequently in adolescents for both types, and Type 2 adolescents and nonadolescents show greater averages of sleep paralysis than Type 1 age counterparts. Type 2 Adolescents in particular show a greater increase in sleep paralysis occurrence compared to Type 1 adolescents. In addition, adolescents experience a slightly greater average of hallucinations and lucid dreams. In Type 1 patients, adolescents experience slightly greater cataplexy averages, but Type 2 nonadolescents experience a slightly greater average. Also, Type 2 patients show greater RBD averages than Type 1 patients, and there is an especially large difference between Type 1 and 2 adolescents. Lucid dreaming averages are greater in adolescent patients, but Type 1 patients of all ages show the greatest average of lucid dreaming.

Regarding SOREM periods, Type 1 adolescent distribution is more random while Type 1 nonadolescent distribution is skewed to the right. Type 2 adolescents show a somewhat normal SOREM distribution while Type 2 nonadolescents show a similar distribution to their nonadolescent Type 1 counterparts. However, this could be due to a lack of available data.

Sleep latency for each age group is not correlated in the slightest.

Interpretation: On average, more adolescents appear to experience sleep paralysis, hallucinations, and lucid dreaming than nonadolescents. These variables could thus be attributed to their still-changing brains. But, sleep paralysis, RBD, and lucid dreaming also seem to be type-dependent (Type 2 patients show greater sleep paralysis and RBD averages, and Type 1 patients show greater lucid dreaming averages). For lucid dreaming and RBD occurrence, type and age appear to play a factor in the averages. Adolescents with Type 1 narcolepsy show far less RBD averages than Type 2, but there is not a major RBD difference between type for nonadolescents. Based on the cataplexy observations, it may be worth looking into whether cataplexy is type AND age dependent. What if the incidence of cataplexy diminishes over time in Type 1 cases but increases in Type 2 patients? This question could only be answered by knowing the onset of narcolepsy symptoms. If cataplexy DOES change over time depending on type, this finding would be highly influential as it would help explain the cause of each type.

It is interesting that Type 1 adolescents show a more random distribution in SOREM period compared to their older Type 1 counterparts. This could possibly be due to their younger, neuroplastic brains or the nature of narcolepsy's onset. Could SOREM occurrence become more predictable/organized years after initial disease onset? This may be possible because nonadolescents show a similar SOREM distribution regardless of type.

```
[5]: #ample data manipulations must be done to find the age group means for each
      ↪categorical variable and narcolepsy type

      #in original dataset, select for desired columns
      agesymp = TypeData[['Age', 'Narcolepsy Type', 'Cataplexy', 'Hypnagogic',
      ↪hallucinations', 'Sleep paralysis', 'RBD (REM behaviour disorder)', 'Lucid
      ↪dreaming']]].fillna(0)
```

```

#make 4 new tables isolated by narcolepsy type and age group (18-25 years is
↳considered adolescence) and save in namespace
young1 = agesymp[(agesymp['Narcolepsy Type'] == '1') & (agesymp['Age'] < 26)]
young2 = agesymp[(agesymp['Narcolepsy Type'] == '2') & (agesymp['Age'] < 26)]
old1 = agesymp[(agesymp['Narcolepsy Type'] == '1') & (agesymp['Age'] > 25)]
old2 = agesymp[(agesymp['Narcolepsy Type'] == '2') & (agesymp['Age'] > 25)]

#in each of the 4 tables, rename the columns and save in namespace
young1.set_axis(['Age', 'Narcolepsy Type', 'Cataplexy', 'Hallucinations',
↳'Sleep paralysis', 'RBD', 'Lucid dreams'], axis='columns', inplace=True)
young2.set_axis(['Age', 'Narcolepsy Type', 'Cataplexy', 'Hallucinations',
↳'Sleep paralysis', 'RBD', 'Lucid dreams'], axis='columns', inplace=True)
old1.set_axis(['Age', 'Narcolepsy Type', 'Cataplexy', 'Hallucinations', 'Sleep
↳paralysis', 'RBD', 'Lucid dreams'], axis='columns', inplace=True)
old2.set_axis(['Age', 'Narcolepsy Type', 'Cataplexy', 'Hallucinations', 'Sleep
↳paralysis', 'RBD', 'Lucid dreams'], axis='columns', inplace=True)

#in each of the 4 tables, select for desired columns, find the mean for each
↳column, convert to dataframe, and rename the resulting column
young_1 = young1[['Cataplexy', 'Hallucinations', 'Sleep paralysis', 'RBD',
↳'Lucid dreams']].mean()
young1data = pd.DataFrame(young_1)
young1data.set_axis(['Type 1 Adolescent'], axis='columns', inplace=True)

young_2 = young2[['Cataplexy', 'Hallucinations', 'Sleep paralysis', 'RBD',
↳'Lucid dreams']].mean()
young2data = pd.DataFrame(young_2)
young2data.set_axis(['Type 2 Adolescent'], axis='columns', inplace=True)

old_1 = old1[['Cataplexy', 'Hallucinations', 'Sleep paralysis', 'RBD', 'Lucid
↳dreams']].mean()
old1data = pd.DataFrame(old_1)
old1data.set_axis(['Type 1 Older'], axis='columns', inplace=True)

old_2 = old2[['Cataplexy', 'Hallucinations', 'Sleep paralysis', 'RBD', 'Lucid
↳dreams']].mean()
old2data = pd.DataFrame(old_2)
old2data.set_axis(['Type 2 Older'], axis='columns', inplace=True)

#merge the 4 tables using the concat function and save in namespace
agetypesymp = pd.concat((young1data, old1data, young2data, old2data), axis=1)

#create a bar graph for the above table with appropriate sizing, axis lables,
↳title, and colors

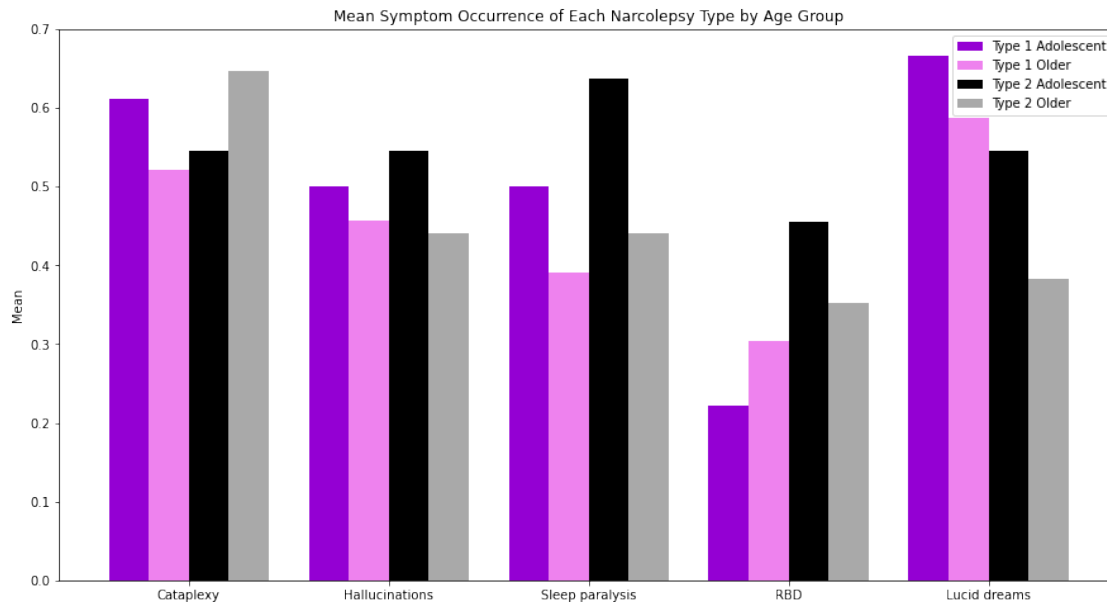
```

```

agetypesymp.plot.bar(figsize=(15,8), rot=0, color=['darkviolet', 'violet', 'k', 'darkgrey'], ylabel='Mean', title='Mean Symptom Occurrence of Each Narcolepsy Type by Age Group', width=0.8)

```

[5]: <AxesSubplot:title={'center': 'Mean Symptom Occurrence of Each Narcolepsy Type by Age Group'}, ylabel='Mean'>



[]:

[6]: #for the numerical data, the steps are similar to those used for sex and SOREM/
↳sleep latency

```

#in original dataset, select for desired columns save in namespace
Agroupsoirem = Typedata[['Age', 'Group', 'Narcolepsy Type', 'Sleep onset in',
↳REM periods']]

#isolate by age group and remove narcolepsy types that are not equal to '1' or
↳'2', save age groups in namespace
adolescent = Agroupsoirem[(Agroupsoirem['Age'] < 25) &
↳(Agroupsoirem['Narcolepsy Type'] > '0')]
Old = Agroupsoirem[(Agroupsoirem['Age'] > 26) & (Agroupsoirem['Narcolepsy',
↳Type'] > '0')]

#rename the columns in each age group table for clarification in the following
↳grid histograms
adolescent.set_axis(['Age', '25 and Under', 'Narcolepsy Type (Adolescents)',
↳'SOREM Periods'], axis='columns', inplace=True)

```

```

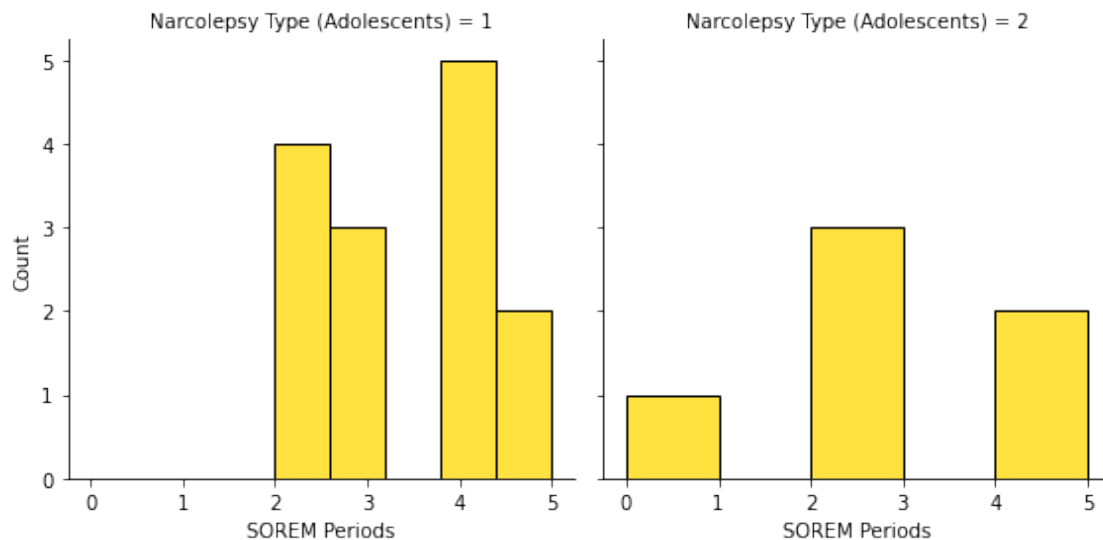
Old.set_axis(['Age', 'Over 25', 'Narcolepsy Type (> 25)', 'SOREM Periods'],
    ↪axis='columns', inplace=True)
#reorder to values for the narcolepsy type column so that the types appear in
    ↪numerical order in the nonadolescent grid
old = Old.sort_values(['Narcolepsy Type (> 25)'])

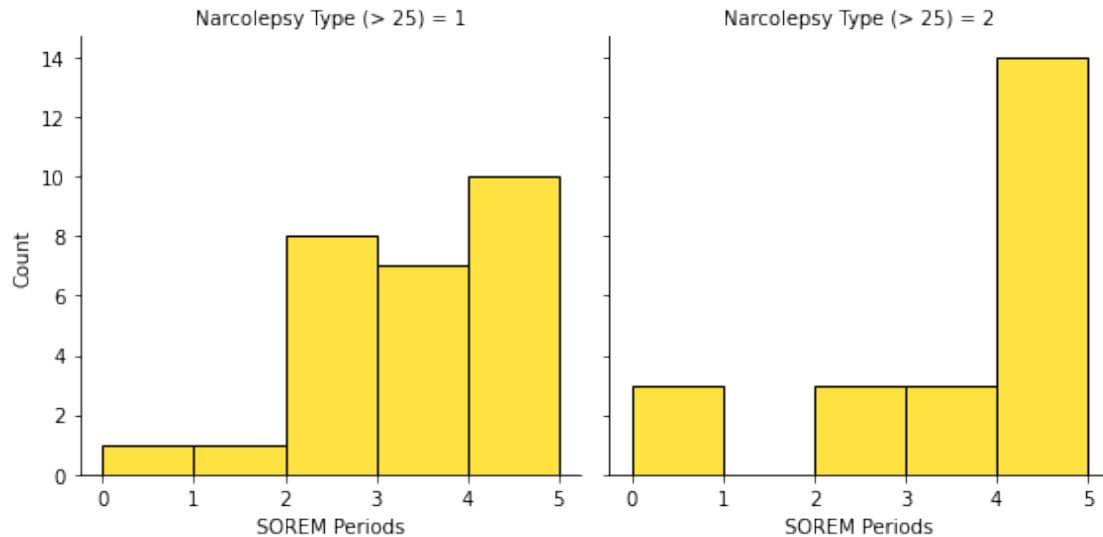
#use seaborn to set up a graphing grid for each narcolepsy type in the younger
    ↪group
grid = sns.FacetGrid(data=adolescent, col='Narcolepsy Type (Adolescents)',
    ↪margin_titles=True, height=4, aspect=1.0)
#use seaborn to make a grid of histograms with SOREM Periods on the x-axis
grid.map_dataframe(sns.histplot, 'SOREM Periods', bins=5, color='gold')

#use seaborn to set up a graphing grid for each narcolepsy type in the
    ↪nonadolescent group
grid = sns.FacetGrid(data=old, col='Narcolepsy Type (> 25)',
    ↪margin_titles=True, height=4, aspect=1.0)
#use seaborn to make a grid of histograms with SOREM Periods on the x-axis
grid.map_dataframe(sns.histplot, 'SOREM Periods', bins=5, color='gold')

```

[6]: <seaborn.axisgrid.FacetGrid at 0x7fb220e9b4f0>





[7]: *#for the numerical variable sleep latency, I chose to view each age group as a*
↳scatterplot to check for linear regression possibilities

```
#in original dataset, select for desired columns, remove NaN values, and save  

↳in namespace  

age_latency = TypeData[['Age', 'Sleep latency (min)']].dropna()  

#remove sleep latencies of zero, isolate each group by specified age range,  

↳save the two groups in namespace  

agelatencyad = age_latency[(age_latency['Sleep latency (min)'] > 0.0) &  

    ↳(age_latency['Age'] < 26)]  

agelatencyold = age_latency[(age_latency['Sleep latency (min)'] > 0.0) &  

    ↳(age_latency['Age'] > 25)]  
  

#plot the two age groups with their associated sleep latencies with age on the  

↳x-axis and sleep latency on the y-axis  

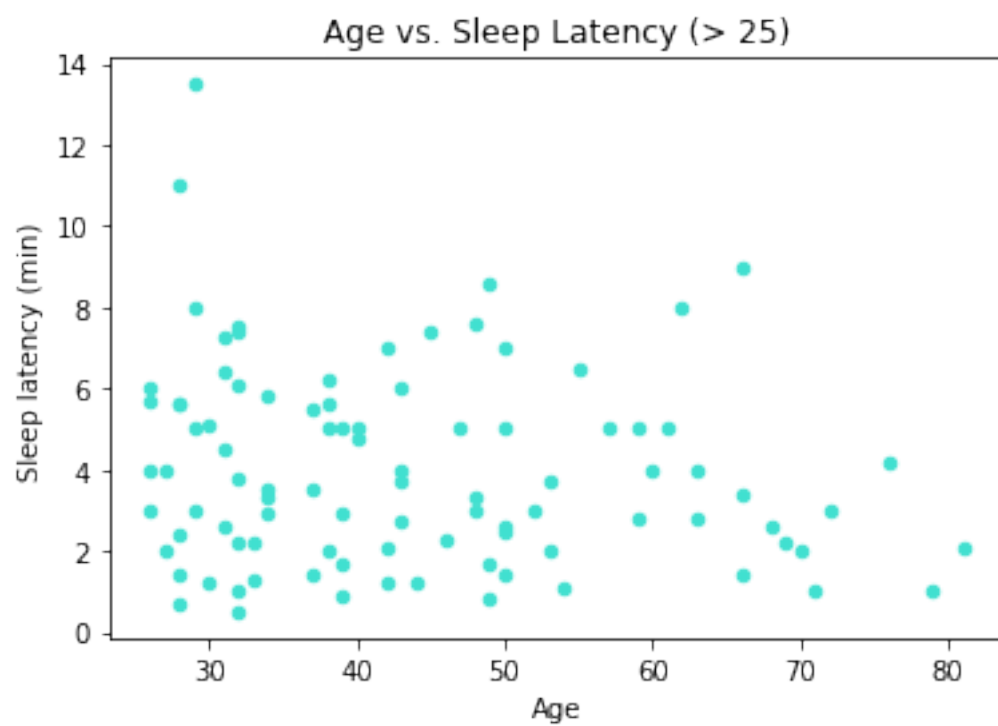
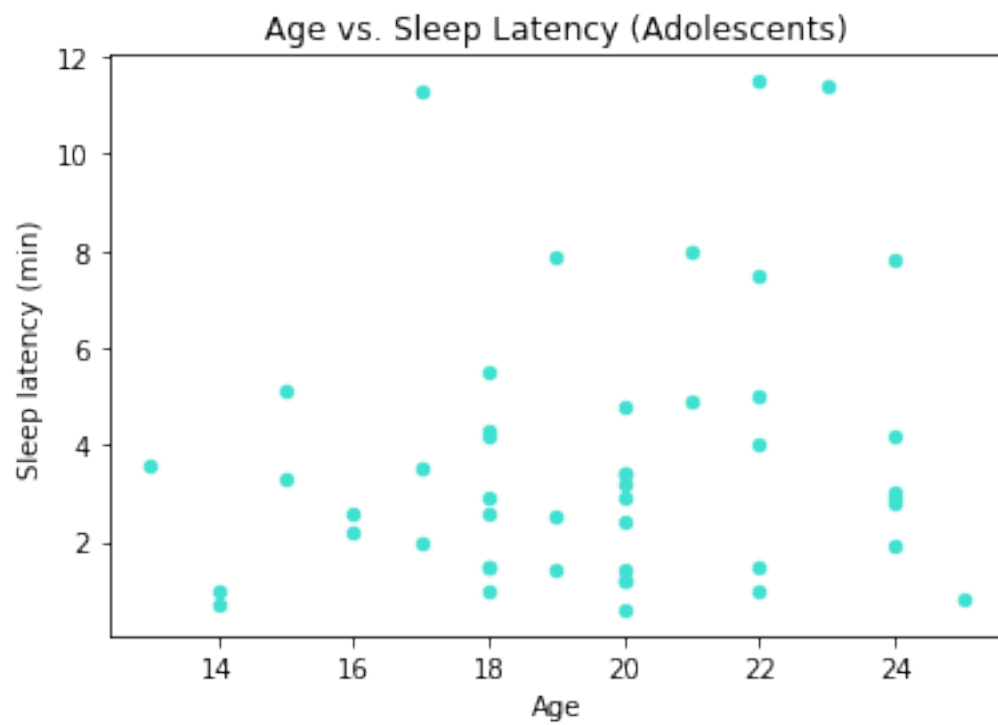
agelatencyad.plot.scatter(x='Age', y='Sleep latency (min)', title='Age vs.  

    ↳Sleep Latency (Adolescents)', color='turquoise')  

agelatencyold.plot.scatter(x='Age', y='Sleep latency (min)', title='Age vs.  

    ↳Sleep Latency (> 25)', color='turquoise')
```

[7]: `<AxesSubplot:title={'center':'Age vs. Sleep Latency (> 25)'}, xlabel='Age',
ylabel='Sleep latency (min)'`



Question 3 - Are there any significant relationships between narcolepsy type and one symptom?

The following outputs in this section are: 1) table of significant dependent relationship p-values along with the variables tested 2) grid histograms of SOREM periods for each narcolepsy type and/or occurrence of lucid dreaming 3) grid of pointplots comparing two variables for each narcolepsy type that were found with significant dependent relationships 4) bar graph showing average occurrence of categorical narcolepsy symptoms for each narcolepsy type

Results:

The following variables for each type were found to have relational significance using the chi-squared test (the p-values are shown in the table): - Type vs. Lucid dreams - Type vs. SOREM - Cataplexy vs. RBD (Type 1) - Cataplexy vs. Hallucinations (Type 1, Type 2) - Cataplexy vs. Sleep paralysis (Type 1) - Cataplexy vs. Lucid dreams (Type 1) - Hallucinations vs. Sleep paralysis (Type 1, Type 2) - Hallucinations vs. RBD (Type 1, Type 2) - Hallucinations vs. Lucid dreams (Type 1) - Sleep paralysis vs. Lucid dreams (Type 1) - RBD vs. Lucid dreams (Type 1) - Sleep paralysis vs. RBD (Type 1, Type 2)

The two variables tested against narcolepsy type shown above did not have statistically significant p-values, but they had significantly lower p-values than other variables tested against narcolepsy type and should be examined further in other populations. The grid histograms attempt to examine this relational possibility further, and it can be observed that there is a clear distributional difference in SOREM periods between type AND the presence of lucid dreams. Type 1 SOREMs alongside lucid dreams and Type 2 SOREMs without lucid dreams show similar, somewhat normal, left-skewed distributions. Upon only observing type and SOREMs in the following histograms, Type 1 has less variation and skewness in SOREM periods than Type 2.

The pointplots allow for easy visualization of the Type 1 and Type 2 significantly dependent variables. The red plots show shared significance between both types while the blue and magenta plots show significance in Type 1. The greater the slope of the connecting line, the greater power the p-value holds for that relationship. The magenta plots indicate dependent relationships involving cataplexy in only the Type 1 group. The blue plots indicate dependent relationships involving lucid dreaming in only the Type 1 group. The Type 1 slopes in the red and blue plots are generally steeper than the Type 2 slopes, showing a clear significant difference in that relationship between types.

The bar graph supports a clear difference in significance of lucid dreaming between types with Type 1 having the greatest average. There are smaller mean differences regardless of other variables for RBD, sleep paralysis, and cataplexy between types. Note that there is a greater cataplexy average for Type 2 participants, even though Type 2 narcolepsy is referred to as “narcolepsy without cataplexy.”

Interpretation: The reason I chose to test the relationships between two variables for each type is to see if any variables could be explained by genetics. For instance, the observation that there are no unique dependent relationships for Type 2 and 6 unique relationships for Type 1 could support previous evidence of multiple heritable genes at work in Type 1 patients. However, this does not rule out heritability in Type 2 patients. Based on the SOREM and lucid dream histograms, it is possible to guess that narcolepsy type and the presence of lucid dreaming could impact SOREM periods in some way. In just the SOREM vs. type histograms, the distributions appear very similar to the nonadolescent distributions for both types. This supports the idea that age influences

SOREMs for both types because the distributions are vastly different for younger narcoleptics.

The pointplots provide quite a bit of interesting information. Firstly, hallucinations and cataplexy, sleep paralysis, and RBD are dependent variables found in both types, as well as RBD and sleep paralysis. These type-independent relationships involving hallucinations fit well with the bar graph in that there is little to no incident difference between types; however, it is intriguing that cataplexy should be tied in with this variable. The type-independent relationship between sleep paralysis and RBD is not surprising considering many REM disorders involve sleep paralysis. Also, sleep paralysis can be classified as an RBD. For the blue pointplots, it is unsurprising that lucid dreaming incidence in Type 2 patients does not change with the addition of another variable because this population shows a much higher overall incidence in Type 2 than Type 1 (which could indicate a neurofunctional or common genetic origin). Although, it is interesting that lucid dreaming incidence can change in Type 1 with the addition of RBDs, hallucinations, and sleep paralysis. This could indicate one or multiple genes are responsible for lucid dreaming as well as any of those other variables. The same could be said for the Type 1 relationships between cataplexy and lucid dreaming, RBDs, and sleep paralysis.

```
[8]: #for this question, I chose to perform chi-square tests on each categorical
      ↪variable to identify dependent relationships
      #after finding the related variables with p-values under 0.05 and 0.1, I
      ↪created pointplots showing the central tendencies of those relationships
      ↪relative to type
      #for the sake of space, I only included the tests with significant or
      ↪interesting p-values

      #in original dataset, select for the desired columns and save in namespace
      allsymptype = TypeData[['Narcolepsy Type', 'Sleep onset in REM periods',
      ↪'Cataplexy', 'Hypnagogic hallucinations', 'Sleep paralysis', 'RBD (REM
      ↪behaviour disorder)', 'Lucid dreaming']]
      #rename the columns in the above new table for easier future manipulations
      allsymptype.set_axis(['Type', 'SOREM', 'Cataplexy', 'HH', 'SP', 'RBD', 'LD'],
      ↪axis='columns', inplace=True)
      #separate the above table into two new tables for each narcolepsy type and save
      ↪in namespace
      type1symp = allsymptype[(allsymptype['Type'] == '1')]
      type2symp = allsymptype[(allsymptype['Type'] == '2')]

      #in the table that was made just before separating by type, the values under
      ↪narcolepsy type are strings
      #we want type values to be integers so that the chi square test can be applied
      ↪to type
      #python will not allow me to replace values in allsymptype, so I had to concat
      ↪the two type tables, then switch the string values with integers using the
      ↪replace function
      alltype = pd.concat((type1symp, type2symp), axis=0)
      alltype['Type'] = alltype['Type'].replace(['1'], 1)
      alltype['Type'] = alltype['Type'].replace(['2'], 2)
```

```

#the following manipulations are chi square tests for two chosen variables in a
↳chosen table
#chi sqr tests are performed using the crosstab function to make a contingency
↳dataframe table (first line) which is saved in the namespace
#the contingency table is then converted to an array which is saved in the
↳namespace (second and third lines)
#using the chi2_contingency() function on the contingency array, the
↳statistical value, p-value, and degree of freedom respectively are found
#the resulting stat value, p, and DF for each test are saved in the namespace
↳so that a table can be made later

#in the table with both narcolepsy types, Type and Lucid dreams are tested
chisqt = pd.crosstab(alltype.Type, alltype.LD, margins=True)
value = np.array([chisqt.iloc[0][0:5].values,
                  chisqt.iloc[1][0:5].values])
c1 = chi2_contingency(value)[0:3]

#in the table with both narcolepsy types, Type and Lucid dreams are tested
chisqt = pd.crosstab(alltype.Type, alltype.SOREM, margins=True)
value = np.array([chisqt.iloc[0][0:5].values,
                  chisqt.iloc[1][0:5].values])
c2 = chi2_contingency(value)[0:3]

#in the table for Type 1, Cataplexy and RBD are tested
chisq = pd.crosstab(typesymp.Cataplexy, typesymp.RBD, margins=True)
value = np.array([chisq.iloc[0][0:5].values,
                  chisq.iloc[1][0:5].values])
c3 = chi2_contingency(value)[0:3]

#in the table for Type 1, Cataplexy and HH are tested
chisq = pd.crosstab(typesymp.Cataplexy, typesymp.HH, margins=True)
value = np.array([chisq.iloc[0][0:5].values,
                  chisq.iloc[1][0:5].values])
c4 = chi2_contingency(value)[0:3]

#in the table for Type 1, Cataplexy and SP are tested
chisq = pd.crosstab(typesymp.Cataplexy, typesymp.SP, margins=True)
value = np.array([chisq.iloc[0][0:5].values,
                  chisq.iloc[1][0:5].values])
c5 = chi2_contingency(value)[0:3]

#in the table for Type 1, Cataplexy and LD are tested
chisq = pd.crosstab(typesymp.Cataplexy, typesymp.LD, margins=True)
value = np.array([chisq.iloc[0][0:5].values,
                  chisq.iloc[1][0:5].values])

```

```

c6 = chi2_contingency(value)[0:3]

#in the table for Type 1, HH and SP are tested
chisq = pd.crosstab(type1symp.HH, type1symp.SP, margins=True)
value = np.array([chisq.iloc[0][0:5].values,
                  chisq.iloc[1][0:5].values])
c7 = chi2_contingency(value)[0:3]

#in the table for Type 1, HH and RBD are tested
chisq = pd.crosstab(type1symp.HH, type1symp.RBD, margins=True)
value = np.array([chisq.iloc[0][0:5].values,
                  chisq.iloc[1][0:5].values])
c8 = chi2_contingency(value)[0:3]

#in the table for Type 1, HH and LD are tested
chisq = pd.crosstab(type1symp.HH, type1symp.LD, margins=True)
value = np.array([chisq.iloc[0][0:5].values,
                  chisq.iloc[1][0:5].values])
c9 = chi2_contingency(value)[0:3]

#in the table for Type 1, LD and SP are tested
chisq = pd.crosstab(type1symp.LD, type1symp.SP, margins=True)
value = np.array([chisq.iloc[0][0:5].values,
                  chisq.iloc[1][0:5].values])
c10 = chi2_contingency(value)[0:3]

#in the table for Type 1, LD and RBD are tested
chisq = pd.crosstab(type1symp.LD, type1symp.RBD, margins=True)
value = np.array([chisq.iloc[0][0:5].values,
                  chisq.iloc[1][0:5].values])
c11 = chi2_contingency(value)[0:3]

#in the table for Type 1, SP and RBD are tested
chisq = pd.crosstab(type1symp.SP, type1symp.RBD, margins=True)
value = np.array([chisq.iloc[0][0:5].values,
                  chisq.iloc[1][0:5].values])
c12 = chi2_contingency(value)[0:3]

#in the table for Type 2, HH and Cataplexy are tested
chisq2 = pd.crosstab(type2symp.HH, type2symp.Cataplexy, margins=True)
value = np.array([chisq2.iloc[0][0:5].values,
                  chisq2.iloc[1][0:5].values])
c13 = chi2_contingency(value)[0:3]

#in the table for Type 2, HH and RBD are tested
chisq2 = pd.crosstab(type2symp.HH, type2symp.RBD, margins=True)
value = np.array([chisq2.iloc[0][0:5].values,

```

```

        chisq2.iloc[1][0:5].values])
c14 = chi2_contingency(value)[0:3]

#in the table for Type 2, HH and SP are tested
chisq2 = pd.crosstab(type2symp.HH, type2symp.SP, margins=True)
value = np.array([chisq2.iloc[0][0:5].values,
                  chisq2.iloc[1][0:5].values])
c15 = chi2_contingency(value)[0:3]

#in the table for Type 2, SP and RBD are tested
chisq2 = pd.crosstab(type2symp.SP, type2symp.RBD, margins=True)
value = np.array([chisq2.iloc[0][0:5].values,
                  chisq2.iloc[1][0:5].values])
c16 = chi2_contingency(value)[0:3]

```

```

[9]: #to make the table of significant results, make a list of the combined results
      ↳and save in namespace
str = (c1, c2, c3, c4, c5, c6, c7, c8, c9, c10, c11, c12, c13, c14, c15, c16)
#convert the list to a dataframe
strdata = pd.DataFrame(str)
#change the column names to reflect the output type
strdata.set_axis(['Stat', 'P', 'DF'], axis='columns', inplace=True)
#change the row names to reflect the relationships tested
strdata.set_axis(['Type, LD', 'Type, SOREM', 'Cataplexy, RBD', 'Cataplexy, HH',
↳'Cataplexy, SP', 'Cataplexy, LD', 'HH, SP', 'HH, RBD', 'HH, LD', 'LD, SP',
↳'LD, RBD', 'SP, RBD', 'Cataplexy, HH2', 'HH, RBD2', 'HH, SP2', 'SP, RBD2'],
↳axis='rows', inplace=True)
strdata

```

```

[9]:
      Stat      P  DF
Type, LD    3.717178  0.155892  2
Type, SOREM  6.131520  0.189539  4
Cataplexy, RBD  8.747160  0.012606  2
Cataplexy, HH   5.343286  0.069139  2
Cataplexy, SP  13.530393  0.001153  2
Cataplexy, LD  11.791392  0.002751  2
HH, SP       27.525188  0.000001  2
HH, RBD      6.958114  0.030836  2
HH, LD      20.037502  0.000045  2
LD, SP      19.660046  0.000054  2
LD, RBD      5.577158  0.061509  2
SP, RBD      9.960811  0.006871  2
Cataplexy, HH2  5.876538  0.052957  2
HH, RBD2     6.281700  0.043246  2
HH, SP2     16.200064  0.000304  2
SP, RBD2     5.148387  0.076215  2

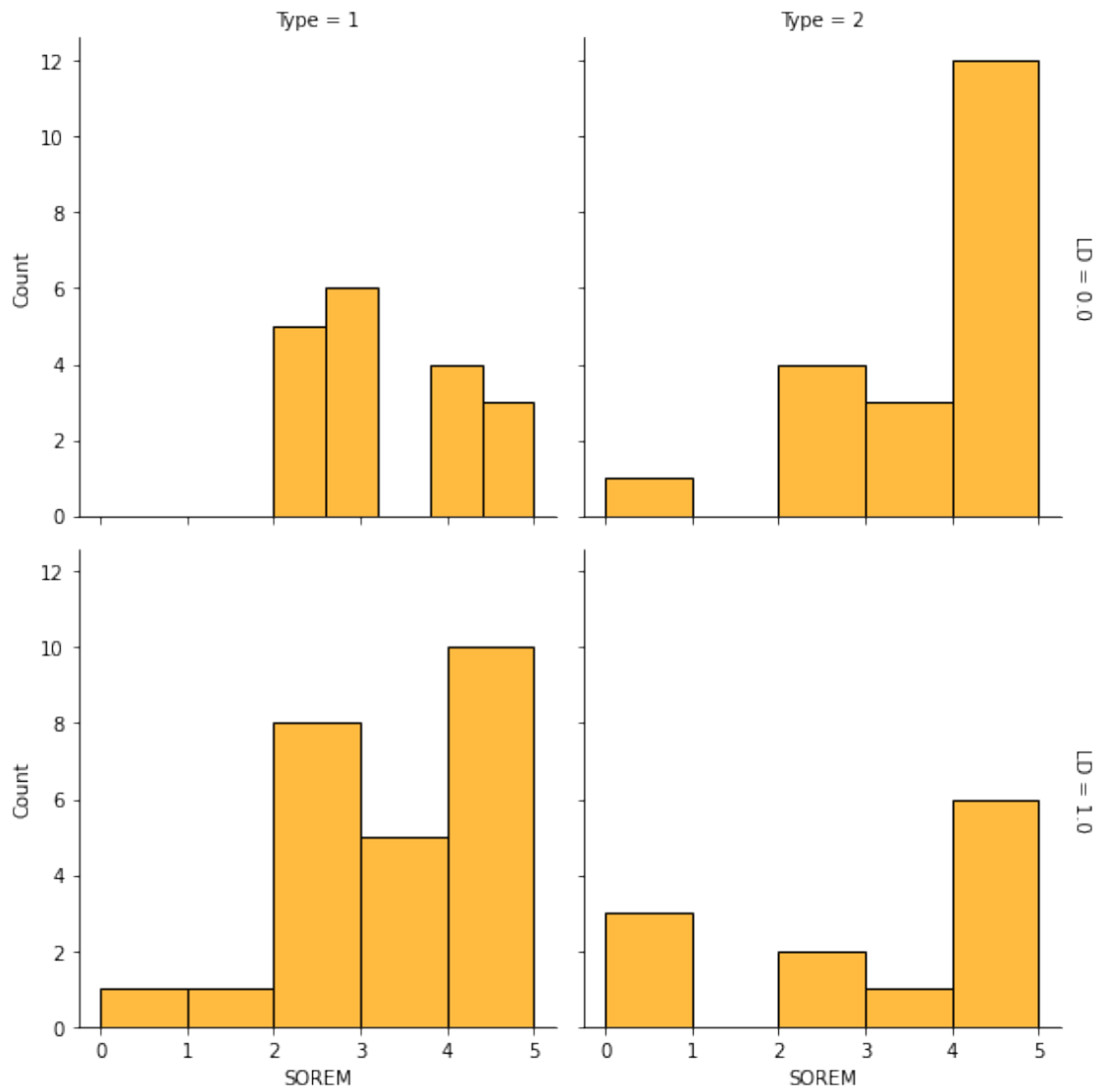
```

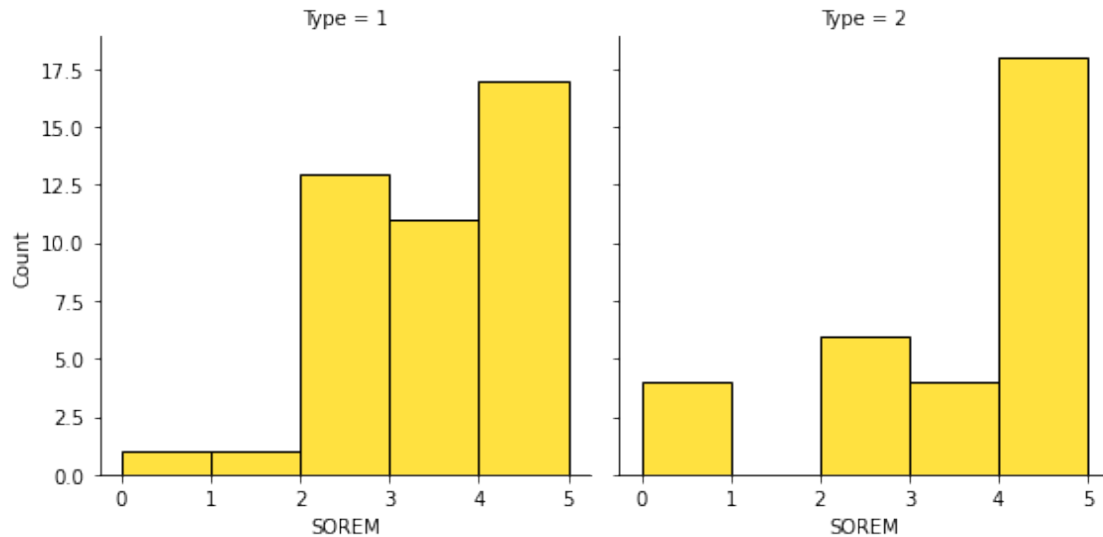
```
[10]: #first, let's visualize the slightly insignificant results from the table with
      ↪ both types
      #these were the smallest resulting p-values found with the rest of the p-values
      ↪ being much larger, which is why they are being included

      #use seaborn to set up a graphing grid for each narcolepsy type and the LD
      ↪ variable
      grid = sns.FacetGrid(data=alltype, row='LD', col='Type', margin_titles=True,
      ↪ height=4, aspect=1.0)
      #use seaborn to make a grid of histograms with SOREM Periods on the x-axis
      grid.map_dataframe(sns.histplot, 'SOREM', bins=5, color='orange')

      #use seaborn to set up a graphing grid for each narcolepsy type
      grid = sns.FacetGrid(data=alltype, col='Type', margin_titles=True, height=4,
      ↪ aspect=1.0)
      #use seaborn to make a grid of histograms with SOREM Periods on the x-axis
      grid.map_dataframe(sns.histplot, 'SOREM', bins=5, color='gold')
```

```
[10]: <seaborn.axisgrid.FacetGrid at 0x7fb0b9f341f0>
```





```
[11]: #to visualize the significant results, many pointplots can be made for each
      ↪variable tested and compared by type
      #these pairplots only show the results from the Type 1 and Type 2 tables with
      ↪red representing dependent relationships found in both types

      #first, a grid of figures is made - since there are 10 unique variable test
      ↪results, the grid needs to be a 2x5 for all 10 subplots
      figure, axes = plt.subplots(2, 5, sharex=True, figsize=(22,6))
      #an overall title is added that sums up what each subplot means
      figure.suptitle('Individual Comparisons of Significant Variables')

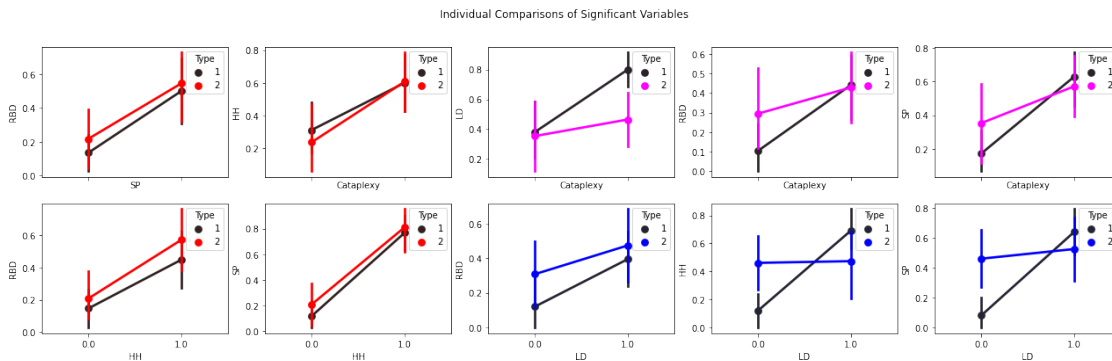
      #using seaborn, individual pointplots are added with specific placement (ax),
      ↪colors for simplification, and legends for each type
      sns.pointplot(ax=axes[0, 4], data=alltype, x='Cataplexy', y='SP', hue='Type',
      ↪kind='point', color='magenta')
      sns.pointplot(ax=axes[0, 1], data=alltype, x='Cataplexy', y='HH', hue='Type',
      ↪kind='point', color='red')
      sns.pointplot(ax=axes[0, 2], data=alltype, x='Cataplexy', y='LD', hue='Type',
      ↪kind='point', color='magenta')
      sns.pointplot(ax=axes[0, 3], data=alltype, x='Cataplexy', y='RBD', hue='Type',
      ↪kind='point', color='magenta')
      sns.pointplot(ax=axes[1, 1], data=alltype, x='HH', y='SP', hue='Type',
      ↪kind='point', color='red')
      sns.pointplot(ax=axes[1, 3], data=alltype, x='LD', y='HH', hue='Type',
      ↪kind='point', color='blue')
      sns.pointplot(ax=axes[1, 0], data=alltype, x='HH', y='RBD', hue='Type',
      ↪kind='point', color='red')
```

```

sns.pointplot(ax=axes[1, 4], data=alltype, x='LD', y='SP', hue='Type',
    ↪kind='point', color='blue')
sns.pointplot(ax=axes[1, 2], data=alltype, x='LD', y='RBD', hue='Type',
    ↪kind='point', color='blue')
sns.pointplot(ax=axes[0, 0], data=alltype, x='SP', y='RBD', hue='Type',
    ↪kind='point', color='red')

```

[11]: <AxesSubplot:xlabel='SP', ylabel='RBD'>



```

[12]: #finally, we will visualize the mean of each categorical symptom by type

#from the type 1 and 2 tables previously made, select for the categorical
    ↪variables and save each new table in namespace
type1mean = type1symp[['Cataplexy', 'HH', 'SP', 'RBD', 'LD']].mean(axis=0)
type2mean = type2symp[['Cataplexy', 'HH', 'SP', 'RBD', 'LD']].mean(axis=0)

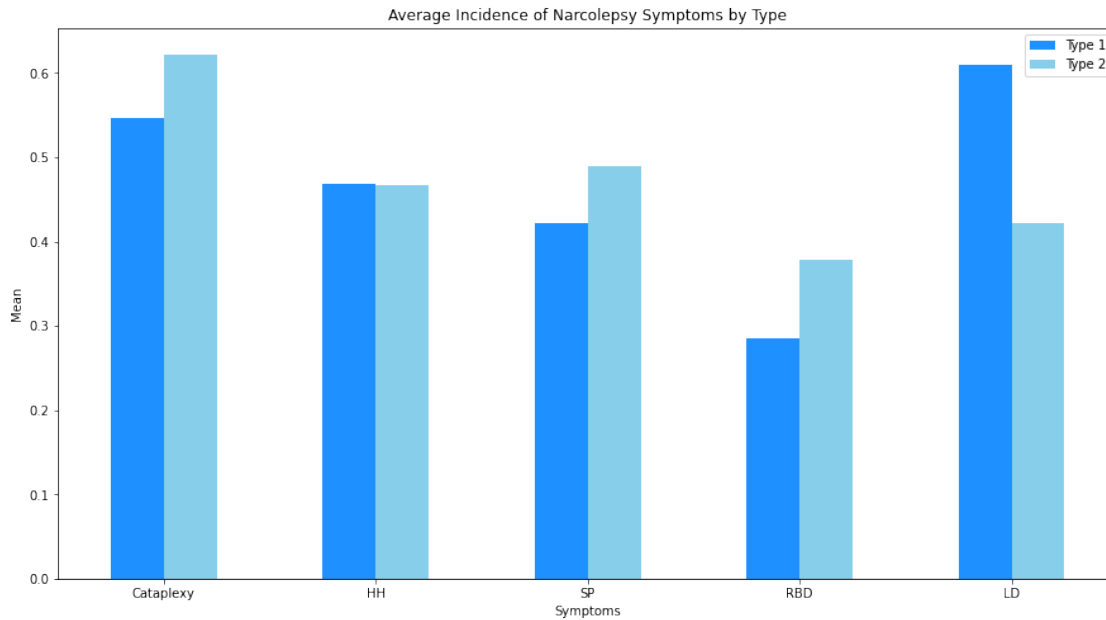
#convert each new table to a dataframe and save in namespace
type1Mean = pd.DataFrame(type1mean)
type2Mean = pd.DataFrame(type2mean)

#merge the two tables using the concat function with the rows as the symptoms
    ↪and columns as type
typemean = pd.concat((type1Mean, type2Mean), axis=1)
#rename the columns for simplicity
typemean.set_axis(['Type 1', 'Type 2'], axis='columns', inplace=True)

#create a bar graph of the merged tables so that symptom incidence within the
    ↪original dataset can be compared by type
typemean.plot.bar(figsize=(15,8), title='Average Incidence of Narcolepsy
    ↪Symptoms by Type', xlabel='Symptoms', ylabel='Mean', color=['dodgerblue',
    ↪'skyblue'], rot=0)

```

```
[12]: <AxesSubplot:title={'center':'Average Incidence of Narcolepsy Symptoms by Type'}, xlabel='Symptoms', ylabel='Mean'>
```



Question 4 - Are there any significant correlations between two symptoms in Group 1 that could be attributed to sex rather than type?

The following outputs are: 1) correlation table for both types 2) correlation table for type 1 3) correlation table for type 2 4) correlation table for both types, both sexes, and all symptoms 5) correlation table for type 1 females 6) correlation table for type 2 females 7) correlation table for type 1 males 8) correlation table for type 2 males

Results: - 0.63 HH vs. SP (Type 1, Type 2, M, F) - 0.66 HH vs. SP (Type 1, M, F) - 0.55 LD vs. SP (Type 1, M, F) - 0.56 HH vs. LD (Type 1, M, F) - 0.60 HH vs. SP (Type 2, M, F) - 0.59 HH vs. SP (Type 1, F) - 0.55 SP vs. LD (Type 1, F) - 0.54 SP vs. LD (Type 1, M) - **0.61 HH vs. LD (Type 1, F)** - **0.53 Cataplexy vs. SP (Type 1, F)** - **0.90 HH vs. SP (Type 2, M)** - **0.57 Cataplexy vs. HH (Type 2, M)** - **0.69 Cataplexy vs. SP (Type 2, M)** - **0.76 HH vs. SP (Type 1, M)**

There are no strong correlations between symptom and type. The unhighlighted correlations show medium correlational significance between variables that are not unique for either sex and/or type. Type 2 participants do not show any significant, unique correlations between variables. From the highlighted correlations, Type 1 females have medium correlations of hallucinations vs. lucid dreaming and cataplexy vs. sleep paralysis. Type 2 males have a medium correlation of cataplexy vs. hallucinations and cataplexy vs. sleep paralysis. Even though significant correlations between hallucinations and sleep paralysis are shared between type and sex, males have the highest correlation (much higher than Type 1 and 2), especially Type 2 males ($r=0.9$). Also, Type 1 females have a medium correlation of cataplexy vs. sleep paralysis. Type 2 females and Type 1 males do not have medium or high correlations between cataplexy and sleep paralysis.

Interpretation:

Since the correlations for hallucinations vs. sleep paralysis are much higher in Type 1 and 2 males compared to Type 1 and 2 females, it is possible that male genetics or sex-specific neurology are involved in this relationship. Could there be a male-specific gene involved in the incidence of both hallucinations and sleep paralysis? Additionally, Type 2 males have a medium correlation between cataplexy and hallucinations, which is not found in Type 1 males or Type 1 and 2 females. So, this relationship must be male-specific, but somehow does not include Type 1 males. Type 1 females show a medium correlation between hallucinations and lucid dreaming, which is much higher than in Type 2 females and Type 1 and 2 males. So, this relationship must be female-specific, but somehow does not include Type 2 females. This is an interesting find because of the greater average of lucid dreaming in the Type 1 participants and how similar the hallucination averages were between types. This could mean that female Type 1 participants with hallucinations are more likely to experience lucid dreaming, which would attribute lucid dreaming as somewhat female-specific. Medium correlations exist for cataplexy vs. sleep paralysis in Type 1 females and Type 2 males, with Type 2 males showing the higher correlation. This eliminates the possibility for sex-specific genes behind this correlation unless there was error in the data regarding cataplexy in Type 2 participants.

To sum up, there is a possible sex-specific reason behind these correlations that are type-independent: hallucinations vs. sleep paralysis (M) ; hallucinations vs. lucid dreaming (F)

```
[13]: #with the significant dependent relationships in mind, we will now examine any
      ↳significant correlations between Type and symptom, and two symptoms within a
      ↳type

#from the original dataset, select for the desired columns
allsymptype = TypeData[['Narcolepsy Type', 'Sleep onset in REM periods',
      ↳'Cataplexy', 'Hypnagogic hallucinations', 'Sleep paralysis', 'RBD (REM
      ↳behaviour disorder)', 'Lucid dreaming']]
#rename the columns for simplicity
allsymptype.set_axis(['Type', 'SOREM', 'Cataplexy', 'HH', 'SP', 'RBD', 'LD'],
      ↳axis='columns', inplace=True)

#to replace str with int in the type column:
#separately select for '1' and '2' values within the type column and save in
      ↳namespace
type1symp = allsymptype[(allsymptype['Type'] == '1')]
type2symp = allsymptype[(allsymptype['Type'] == '2')]
#merge the two previous tables with the concat function and save in namespace
alltype = pd.concat((type1symp, type2symp), axis=0)
#use the replace function to change the str to ints
alltype['Type'] = alltype['Type'].replace(['1'], 1)
alltype['Type'] = alltype['Type'].replace(['2'], 2)

#use the correlation function corr() on each relevant table and use display()
      ↳to see all three correlation tables at once
display(alltype.corr(), type1symp.corr(), type2symp.corr())
```

	Type	SOREM	Cataplexy	HH	SP	RBD	\
Type	1.000000	0.049896	0.075113	-0.002056	0.066327	0.096977	
SOREM	0.049896	1.000000	0.199583	0.065013	0.077498	0.098688	
Cataplexy	0.075113	0.199583	1.000000	0.317298	0.361434	0.276356	
HH	-0.002056	0.065013	0.317298	1.000000	0.631064	0.349003	
SP	0.066327	0.077498	0.361434	0.631064	1.000000	0.376018	
RBD	0.096977	0.098688	0.276356	0.349003	0.376018	1.000000	
LD	-0.184669	-0.159473	0.278362	0.326572	0.329944	0.219057	

	LD
Type	-0.184669
SOREM	-0.159473
Cataplexy	0.278362
HH	0.326572
SP	0.329944
RBD	0.219057
LD	1.000000

	SOREM	Cataplexy	HH	SP	RBD	LD
SOREM	1.000000	0.145829	0.057891	-0.044006	0.086874	-0.096786
Cataplexy	0.145829	1.000000	0.288944	0.459796	0.372618	0.429232
HH	0.057891	0.288944	1.000000	0.655806	0.332335	0.559541
SP	-0.044006	0.459796	0.655806	1.000000	0.397628	0.554246
RBD	0.086874	0.372618	0.332335	0.397628	1.000000	0.297534
LD	-0.096786	0.429232	0.559541	0.554246	0.297534	1.000000

	SOREM	Cataplexy	HH	SP	RBD	LD
SOREM	1.000000	0.253280	0.068657	0.184289	0.104292	-0.213550
Cataplexy	0.253280	1.000000	0.361372	0.211912	0.134454	0.109297
HH	0.068657	0.361372	1.000000	0.600001	0.373622	0.012025
SP	0.184289	0.211912	0.600001	1.000000	0.338243	0.064005
RBD	0.104292	0.134454	0.373622	0.338243	1.000000	0.169101
LD	-0.213550	0.109297	0.012025	0.064005	0.169101	1.000000

```
[14]: #based on my personal curiosity, I wanted to see if any correlations grew
      ↪stronger/weaker when controlling for sex

      #from the original dataset, make 4 new tables selecting for sex and narcolepsy
      ↪type, and save each in namespace
      fnarc1 = TypeData[(TypeData['Sex'] == 'F') & (TypeData['Narcolepsy Type'] ==
      ↪'1')]
      mnarc1 = TypeData[(TypeData['Sex'] == 'M') & (TypeData['Narcolepsy Type'] ==
      ↪'1')]
      fnarc2 = TypeData[(TypeData['Sex'] == 'F') & (TypeData['Narcolepsy Type'] ==
      ↪'2')]
      mnarc2 = TypeData[(TypeData['Sex'] == 'M') & (TypeData['Narcolepsy Type'] ==
      ↪'2')]
```

```

#select for the same desired columns in all 4 tables
fnarc = fnarc1[['Age', 'Narcolepsy Type', 'Hypnagogic hallucinations', 'Lucid_
↳dreaming', 'Sleep paralysis', 'RBD (REM behaviour disorder)', 'Cataplexy']]
f2narc = fnarc2[['Age', 'Narcolepsy Type', 'Hypnagogic hallucinations', 'Lucid_
↳dreaming', 'Sleep paralysis', 'RBD (REM behaviour disorder)', 'Cataplexy']]
m1narc = mnarc1[['Age', 'Narcolepsy Type', 'Hypnagogic hallucinations', 'Lucid_
↳dreaming', 'Sleep paralysis', 'RBD (REM behaviour disorder)', 'Cataplexy']]
m2narc = mnarc2[['Age', 'Narcolepsy Type', 'Hypnagogic hallucinations', 'Lucid_
↳dreaming', 'Sleep paralysis', 'RBD (REM behaviour disorder)', 'Cataplexy']]

#to replace str with int in the type column:
#merge the two previous tables with the concat function and save in namespace
sexalltype = pd.concat((fnarc, f2narc, m1narc, m2narc), axis=0)
#use the replace function to change the strs to ints
sexalltype['Narcolepsy Type'] = sexalltype['Narcolepsy Type'].replace(['1'], 1)
sexalltype['Narcolepsy Type'] = sexalltype['Narcolepsy Type'].replace(['2'], 2)

#use the correlation function for the 4 tables and use display() to see all at_
↳once
display(sexalltype.corr(), fnarc.corr(), f2narc.corr(), m1narc.corr(), m2narc.
↳corr())

```

	Age	Narcolepsy Type \
Age	1.000000	0.034796
Narcolepsy Type	0.034796	1.000000
Hypnagogic hallucinations	-0.094248	-0.002056
Lucid dreaming	-0.088426	-0.184669
Sleep paralysis	-0.152001	0.066327
RBD (REM behaviour disorder)	0.011621	0.096977
Cataplexy	-0.010204	0.075113

	Hypnagogic hallucinations	Lucid dreaming \
Age	-0.094248	-0.088426
Narcolepsy Type	-0.002056	-0.184669
Hypnagogic hallucinations	1.000000	0.326572
Lucid dreaming	0.326572	1.000000
Sleep paralysis	0.631064	0.329944
RBD (REM behaviour disorder)	0.349003	0.219057
Cataplexy	0.317298	0.278362

	Sleep paralysis	RBD (REM behaviour disorder) \
Age	-0.152001	0.011621
Narcolepsy Type	0.066327	0.096977
Hypnagogic hallucinations	0.631064	0.349003
Lucid dreaming	0.329944	0.219057
Sleep paralysis	1.000000	0.376018
RBD (REM behaviour disorder)	0.376018	1.000000

Cataplexy	0.361434	0.276356
-----------	----------	----------

	Cataplexy
Age	-0.010204
Narcolepsy Type	0.075113
Hypnagogic hallucinations	0.317298
Lucid dreaming	0.278362
Sleep paralysis	0.361434
RBD (REM behaviour disorder)	0.276356
Cataplexy	1.000000

	Age	Hypnagogic hallucinations \
Age	1.000000	0.342143
Hypnagogic hallucinations	0.342143	1.000000
Lucid dreaming	0.076774	0.606392
Sleep paralysis	0.130844	0.593846
RBD (REM behaviour disorder)	0.091419	0.314303
Cataplexy	-0.139809	0.327132

	Lucid dreaming	Sleep paralysis \
Age	0.076774	0.130844
Hypnagogic hallucinations	0.606392	0.593846
Lucid dreaming	1.000000	0.550026
Sleep paralysis	0.550026	1.000000
RBD (REM behaviour disorder)	0.204980	0.443991
Cataplexy	0.459893	0.527914

	RBD (REM behaviour disorder)	Cataplexy
Age	0.091419	-0.139809
Hypnagogic hallucinations	0.314303	0.327132
Lucid dreaming	0.204980	0.459893
Sleep paralysis	0.443991	0.527914
RBD (REM behaviour disorder)	1.000000	0.344051
Cataplexy	0.344051	1.000000

	Age	Hypnagogic hallucinations \
Age	1.000000	-0.409597
Hypnagogic hallucinations	-0.409597	1.000000
Lucid dreaming	0.138377	-0.076923
Sleep paralysis	-0.354246	0.384615
RBD (REM behaviour disorder)	-0.102833	0.474342
Cataplexy	0.044970	0.182574

	Lucid dreaming	Sleep paralysis \
Age	1.383773e-01	-0.354246
Hypnagogic hallucinations	-7.692308e-02	0.384615
Lucid dreaming	1.000000e+00	0.076923
Sleep paralysis	7.692308e-02	1.000000
RBD (REM behaviour disorder)	-2.194271e-17	0.316228

Cataplexy	9.374785e-17	-0.182574
-----------	--------------	-----------

	RBD (REM behaviour disorder)	Cataplexy
Age	-1.028331e-01	4.497015e-02
Hypnagogic hallucinations	4.743416e-01	1.825742e-01
Lucid dreaming	-2.194271e-17	9.374785e-17
Sleep paralysis	3.162278e-01	-1.825742e-01
RBD (REM behaviour disorder)	1.000000e+00	5.773503e-02
Cataplexy	5.773503e-02	1.000000e+00

	Age	Hypnagogic hallucinations \
Age	1.000000	-0.072137
Hypnagogic hallucinations	-0.072137	1.000000
Lucid dreaming	-0.245770	0.487446
Sleep paralysis	-0.346513	0.762821
RBD (REM behaviour disorder)	0.082573	0.359546
Cataplexy	0.258408	0.242511

	Lucid dreaming	Sleep paralysis \
Age	-0.245770	-0.346513
Hypnagogic hallucinations	0.487446	0.762821
Lucid dreaming	1.000000	0.542370
Sleep paralysis	0.542370	1.000000
RBD (REM behaviour disorder)	0.385077	0.294174
Cataplexy	0.336123	0.292440

	RBD (REM behaviour disorder)	Cataplexy
Age	0.082573	0.258408
Hypnagogic hallucinations	0.359546	0.242511
Lucid dreaming	0.385077	0.336123
Sleep paralysis	0.294174	0.292440
RBD (REM behaviour disorder)	1.000000	0.327327
Cataplexy	0.327327	1.000000

	Age	Hypnagogic hallucinations \
Age	1.000000	-0.474273
Hypnagogic hallucinations	-0.474273	1.000000
Lucid dreaming	-0.409914	0.108631
Sleep paralysis	-0.227148	0.898933
RBD (REM behaviour disorder)	-0.185926	0.232621
Cataplexy	-0.192254	0.568182

	Lucid dreaming	Sleep paralysis \
Age	-0.409914	-0.227148
Hypnagogic hallucinations	0.108631	0.898933
Lucid dreaming	1.000000	0.035806
Sleep paralysis	0.035806	1.000000
RBD (REM behaviour disorder)	0.420041	0.368035
Cataplexy	0.108631	0.685437

	RBD (REM behaviour disorder)	Cataplexy
Age	-0.185926	-0.192254
Hypnagogic hallucinations	0.232621	0.568182
Lucid dreaming	0.420041	0.108631
Sleep paralysis	0.368035	0.685437
RBD (REM behaviour disorder)	1.000000	0.232621
Cataplexy	0.232621	1.000000

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